Hanh Tran

Curriculum Vitae — Updated July 2026

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EDUCATION

2023–Present Ph.D. in Molecular, Cellular, and Integrative Biosciences (with Graduate Minor in Statistics), Pennsylvania State University, University Park, PA. Advisor: Dr. Emily Davenport.

2018–2020 B.S. in Biology with Honors, Bucknell University, Lewisburg, PA. Honors Thesis: "Identification of odorant receptor genes in the antennal transcriptome of the common Eastern firefly, Photinus pyralis." Advisor: Dr. Sarah Lower.

2015–2018 A.A. in General Studies with Honors, Lehigh Carbon Community College, Schnecksville, PA.

RESEARCH POSITIONS

2023–Present Graduate Research Assistant, Davenport Lab, Pennsylvania State University, University Park, PA.

2022–2023 Research Technologist II, Davenport Lab and Bell Lab, Pennsylvania State University, University Park, PA.

2020–2022 Laboratory Technician, Tekle Lab, Spelman College, Atlanta, GA.

2019–2020 Undergraduate Research Assistant, Lower Lab, Bucknell University, Lewisburg, PA.

HONORS, FELLOWSHIPS, AND AWARDS

2025 Honorable Mention, NSF Graduate Research Fellowship Program.

2024–2026 Predoctoral Training Fellowship, NIH T32 Eukaryotic Gene Regulation Training Program (T32GM152354), Pennsylvania State University.

2023 Homer F. Braddock Scholarship in Biology, Chemistry, and Physics, Pennsylvania State University.

2020 Charlotte Mangum Travel Award for Undergraduate Students, Society of Integrative and Comparative Biology (SICB).

2019 Undergraduate Student Travel Award, ACM Conference on Bioinformatics, Computational, and Health Informatics (ACM-BCB), Niagara Falls, NY.

2018–2020 Bucknell Community College Scholarship, Bucknell University.

2016–2018 Lehigh Carbon Community College Honors Scholarship.

PUBLICATIONS

Preprints

1. **Tran HN**, Kirven KJ, Davenport ER. SCiMS: Sex Calling in Metagenomic Sequences. bioRxiv. 2026 Feb 18. PMCID: PMC12934929.
2. Ryu PR, Keller CA, Nichols RG, **Tran HN**, Brocious PR, Harris LR, Koltun WA, Yochum GS, Davenport ER. Integrative host transcriptomic and mucosal microbiome profiling reveals region-specific host-microbiome associations across the human intestine. bioRxiv. 2026 May 14.

Published

1. Lower SE, Pring S, **Tran H**, Martinson K, Deering S, Price MA, Vestal B, Pask GM, Collins DB, Mitchell RF. Antennal RNAseq reveals odorant receptors with sex-biased expression in the common eastern firefly, *Photinus pyralis*. BMC Genomics. 2025 Jul 14;26(1):651. PMCID: PMC12257845.
2. Chulkina M, **Tran H**, Uribe G, McAninch SB, McAninch C, Seideneck A, He B, Lanza M, Khanipov K, Golovko G, Powell DW, Davenport ER, Pinchuk IV. MyD88-mediated signaling in intestinal fibroblasts regulates macrophage antimicrobial defense and prevents dysbiosis in the gut. Cell Reports. 2025 May 27;44(5):115553. PMCID: PMC12262590.

3. Tekle YI, **Tran H**, Wang F, Singla M, Udu I. Omics of an enigmatic marine amoeba uncovers unprecedented gene trafficking from giant viruses and provides insights into its complex life cycle. Microbiology Research. 2023 Jun;14(2):656–672. PMCID: PMC10521059.

4. Tekle YI, Wang F, **Tran H**, Hayes TD, Ryan JF. The draft genome of *Cochliopodium minus* reveals a complete meiosis toolkit and provides insight into the evolution of sexual mechanisms in Amoebozoa. Scientific Reports. 2022 Jun 14;12(1):9841. PMCID: PMC9198077.

Honors Thesis

1. **Tran HN**. Identification of odorant receptor genes in the antennal transcriptome of the common Eastern firefly, Photinus pyralis. Bucknell University Honors Theses, 532. 2020. https://digitalcommons.bucknell.edu/honors_theses/532.

SCIENTIFIC PRESENTATIONS

2026 Tran H. "Characterizing host gene-fungal crosstalk at the intestinal mucosa." One Health Microbiome Center Symposium. Penn State University.

2025 Tran H. "Exploring the relationship between gut fungi and host gene expression." Center for Eukaryotic Gene Regulation (CEGR) Annual Retreat. Penn State University.

2025 Tran H. " SCiMS: Sex Calling in Metagenomic Sequences." Cold Spring Harbor Laboratory Biology of Genomes.

2024 Tran H. "Exploring the relationship between gut fungi and host gene expression." Center for Eukaryotic Gene Regulation (CEGR) Annual Retreat. Penn State University.

2024 Tran H. " Associations between gut fungi and host gene expression." Cold Spring Harbor Laboratory Microbiome.

2020 Tran H. "Using a computational approach to identify odorant receptor (OR) genes in the North American firefly *Photinus pyralis*." Society of Integrative and Comparative Biology (SICB) Annual Meeting. Poster P1-51. Austin, TX.

TEACHING, MENTORSHIP, AND OUTREACH

Teaching

2023–Present Teaching Assistant (annual), One Health Microbiome Center Kickstart Workshop, Pennsylvania State University. Train incoming graduate students and early-career researchers in microbiome study design and analysis.

2026–Present Organizer and Facilitator, Monthly Trainee Meetings, NIH T32 Eukaryotic Gene Regulation Training Program, Center for Eukaryotic Gene Regulation, Pennsylvania State University.

2022 Instructor, "Introduction to Bioinformatics" computational workshop, Spelman College, Atlanta, GA.

2022 Teaching Assistant, Bioinformatic Research course (Dr. Yonas Tekle), Spelman College, Atlanta, GA.

Undergraduate Mentorship

2025–Present Anwi Tadzong (undergraduate), Davenport Lab, Pennsylvania State University.

2023–2024 Patricia Brocious (undergraduate), Davenport Lab, Pennsylvania State University.

2022–2023 Maya G. Blasingame and Gabrielle S. Scales (undergraduates), Tekle Lab, Spelman College.

Science Outreach

2024 Developer and Workshop Leader, Gut Microbiome Workshop for middle school students, ENVISION: STEM Career Day Supporting Young Women, Pennsylvania State University.

2022 Volunteer, STEM Activities and Workshops for students at Milton Hershey School, organized by Penn State Eberly College of Science Office of Science Outreach.

2019 Volunteer, Pennsylvania Firefly Annual Festival. Demonstrated firefly bioluminescence to children and engaged them with science and conservation.

PROFESSIONAL DEVELOPMENT AND TRAINING

2026 Mendelian Randomization Boot Camp, Skills for Health and Research Professionals (SHARP) Program, Columbia University Mailman School of Public Health (planned, August 2026).

2026 Cell Culture Workshop, Pennsylvania State University (planned, Fall 2026).

RESEARCH SKILLS

Bioinformatic and Statistical Methods: RNA-seq analysis, ITS2 amplicon sequencing analysis, 16S rRNA amplicon sequencing analysis, microbiome community profiling (DADA2, SILVA, ALDEx2), de novo genome and transcriptome assembly, gene annotation, read mapping and alignment, comparative genomics, differential gene expression analysis (DESeq2), weighted gene co-expression network analysis (WGCNA), sparse canonical correlation analysis (sparse CCA), elastic net regression with stability selection, Mendelian randomization, gene set enrichment analysis (GSEA), pathway enrichment analysis (clusterProfiler, MSigDB), phylogenetics, multivariate statistics.

Programming Languages: R, Bash/Unix, Python.

Sequencing Technologies: Illumina, 10X Genomics, PacBio, Oxford Nanopore. Experience with RNA-seq, whole-genome sequencing, metagenomic sequencing, 16S rRNA amplicon sequencing, and ITS2 amplicon sequencing.

Laboratory Methods: DNA and RNA extraction (including from low-biomass mucosal samples), PCR, gel electrophoresis, RNA-FISH, immunocytochemistry, confocal microscopy, bacterial/fungal/amoeba culturing, ITS2 amplicon library preparation.

High-Performance Computing: SLURM workload management on HPC clusters; pipeline development for transcriptomic and microbiome analyses on the Roar Collab cluster (Pennsylvania State University ICDS).

Field Work: Insect sampling, behavioral observation, and dissection of adult fireflies; field-based species identification of fireflies.